

TJAKAAPE, Sweden

WMO: 021410

Lat: 66.300N

Lon: 19.200E

Elev: 1909

StdP: 13.71

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.2	-8.7	-17.8	2.2	-13.0	-12.7	3.0	-8.4	20.4	28.3	18.1	25.7	1.1	20	0.745

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.4	74.1	58.7	70.3	56.8	66.7	55.3	61.4	69.9	59.2	66.5	57.1	63.6	6.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.3	78.1	62.9	56.2	72.1	60.9	54.3	67.3	59.1	28.3	70.4	26.7	66.2	25.3	64.0	69.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.6	15.5	13.7	-19.8	79.7	5.1	4.4	-23.5	82.8	-26.5	85.4	-29.4	87.8	-33.1	91.0		
(5)			-20.3	64.2	4.5	2.9	-23.6	66.2	-26.2	67.9	-28.8	69.5	-32.1	71.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.6	11.8	12.2	19.4	29.1	39.9	49.0	55.5	51.4	42.6	30.2	20.4	16.4
	DBStd	17.24	10.96	11.25	8.94	7.44	7.48	7.00	5.87	5.98	6.00	8.10	8.74	10.01
	HDD50	7083	1186	1057	949	627	327	101	13	53	229	613	888	1040
	HDD65	12193	1651	1477	1414	1077	777	481	302	422	671	1078	1338	1505
	CDD50	375	0	0	0	0	15	72	182	98	8	0	0	0
	CDD65	9	0	0	0	0	0	1	7	1	0	0	0	0
	CDH74	109	0	0	0	0	2	24	74	9	0	0	0	0
	CDH80	10	0	0	0	0	0	2	8	0	0	0	0	0
Wind	WSAvg	6.0	3.4	4.8	7.1	6.9	7.1	7.3	6.7	6.2	6.7	6.2	4.7	4.5
Precipitation	PrecAvg	23.8	1.5	1.1	1.1	1.0	1.8	2.8	3.9	3.0	2.2	1.9	1.8	1.6
	PrecMax	33.3	3.3	2.0	2.2	2.2	3.7	5.0	5.8	5.7	7.4	3.9	3.7	4.9
	PrecMin	14.6	0.0	0.0	0.3	0.3	0.4	0.9	1.1	1.0	0.6	0.6	0.5	0.4
	PrecStd	4.6	0.7	0.6	0.6	0.6	0.9	1.1	1.3	1.4	1.5	0.8	1.0	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	39.1	42.7	54.7	71.3	77.2	80.4	75.1	63.4	49.5	41.6	37.7
		MCWB	37.0	36.5	36.7	43.5	52.2	59.3	63.5	59.8	53.5	45.5	40.3	36.0
	2%	DB	33.9	35.3	37.9	48.4	65.0	71.2	75.7	71.0	58.0	46.3	38.2	34.8
		MCWB	32.2	32.8	33.7	39.3	50.5	55.4	59.6	58.5	51.5	43.6	37.2	33.4
	5%	DB	29.9	31.5	35.1	43.7	58.3	66.9	71.8	67.1	54.6	43.9	34.4	32.0
		MCWB	29.1	29.9	31.8	37.1	47.7	53.3	58.8	56.8	49.4	42.0	33.7	31.5
	10%	DB	26.0	28.2	32.5	40.5	53.0	62.4	67.8	62.9	52.1	41.2	31.9	29.5
		MCWB	25.7	27.4	30.1	35.3	44.7	51.8	56.6	55.3	48.4	39.9	31.6	29.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.7	36.6	36.8	43.7	54.6	61.7	66.0	63.1	55.1	47.0	40.4	36.4
		MCDB	39.0	38.7	40.5	53.5	67.2	72.0	74.4	70.2	57.5	47.9	40.8	37.4
	2%	WB	32.7	33.1	34.2	40.4	51.7	58.2	62.3	60.3	52.7	44.1	37.3	33.5
		MCDB	33.7	34.6	37.2	46.5	61.5	66.7	72.2	66.6	56.6	45.5	37.9	34.4
	5%	WB	29.1	30.1	32.4	37.9	48.2	55.4	60.1	58.4	50.9	41.9	33.8	31.4
		MCDB	29.9	31.2	34.8	42.4	58.0	63.3	68.5	64.4	53.6	43.6	34.3	32.0
	10%	WB	25.7	27.2	30.3	35.8	45.1	52.8	57.9	56.4	49.2	39.9	31.7	29.2
		MCDB	26.1	28.2	32.4	40.4	51.8	60.8	65.1	61.8	51.2	41.2	32.0	29.5
Mean Daily Temperature Range	5% DB	MDBR	11.5	12.7	14.6	15.0	15.2	16.1	16.4	16.0	12.9	10.1	9.8	10.4
		MCDBR	11.5	11.7	13.9	18.6	24.8	22.9	23.3	23.0	18.1	11.8	8.7	10.0
		MCWBR	11.1	10.7	10.9	11.5	13.3	10.8	10.6	11.5	9.3	8.5	9.7	
	5% WB	MCDBR	11.2	12.0	13.3	16.2	23.7	19.7	20.2	19.0	14.0	10.8	8.6	9.9
		MCWBR	10.9	11.3	11.0	10.4	13.1	10.4	10.3	10.7	9.6	8.8	8.4	9.7
Clear-Sky Solar Irradiance	taub	0.172	0.240	0.273	0.301	0.318	0.320	0.330	0.315	0.287	0.265	0.340	0.175	
	taud	1.939	2.174	2.245	2.305	2.488	2.527	2.509	2.535	2.592	2.416	2.838	1.954	
	Ebn at Noon	86	202	253	274	282	283	276	270	255	196	92	10	
	Edh at Noon	5	16	25	30	29	28	28	24	18	13	5	1	
All-Sky Solar Radiation	RadAvg	25	191	652	1232	1560	1622	1493	1119	625	226	48	3	
	RadStd	2	29	41	65	156	148	168	112	82	36	6	0	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (39 neighbors)		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page